

■ A.5 Setting up Mathematical Databases

■ A.5.1 Units Conversions

By giving a sequence of definitions for symbolic objects, you can effectively set up a “database” of information in *Mathematica*. Here is part of a database of units, together with a function `Convert` that converts between different units. Notice that `Convert` prints a message if the units are not compatible.

```
Convert::usage =
"Convert[old,new] converts old to a form involving the combination of units new."
Convert[old_, new_] :=
  Block[{t},
    t = t /. $ToSI ;
    t = N[t] ;
    If[!NumberQ[t], Message[Convert::incomp, old, new]; Return[old],
      Return[t new]
    ]
  ]

Convert::incomp = "Incompatible units in `1` and `2`."

$ToSI =
{
  Inch -> 2.54*10^-2 Meter,
  Foot -> 30.48*10^-2 Meter,
  Mile -> 1.609344*10^3 Meter,
  Minute -> 60 Second,
  Hour -> 60 Minute,
  Day -> 24 Hour,
  Year -> 365 Day
}
```

Part of a package that does units conversions.

Read in the units conversion package. `In[1]:= <<Units.m`

This converts 6 miles per day into meters per second. `In[2]:= Convert[6 Mile/Day, Meter/Second]`

$$\text{Out[2]} = \frac{0.11176 \text{ Meter}}{\text{Second}}$$

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If you try to convert between incompatible units, the package prints a message.

```
In[3]:= Convert[6 Year, Inch]
Convert::incomp: Incompatible units in 6 Year and Inch.
Out[3]= 6 Year
```